

.NET Full-Stack Best Practices and Strategies for IT Professionals

The goal is to create enough articles covering key points regarding best practices and strategies for Full-Stack developing in .NET to then gather and extend their content in a book or guide for IT professionals who want to dig in the topics below. Additionally, quality source code and specific project or solution related technical documentation will be left available as back up to the theory.

The next list provides the links of the most significant projects up today for each topic. The ones that are marked in yellow have their correspondent article.

Articles GitHub folder: <https://github.com/kraftcoding/IT-Document-Collection/tree/master/Kraftcoding%20Articles>

Topics:

- 1 Blazor ([Blazor-SignalR-Chat-.NET-8](#))
- 2 Web APIs ([Book Store v1.x](#))
- 3 Front-End Technologies
 - 3.1 Angular ([Book Store v1.x](#))
 - 3.2 React ([Book Store v2.x](#))
- 4 Authentication and Security in IT projects
 - 4.1 [Implementing JWT Web API security in .NET](#) ([Book Store v1.0](#))
 - 4.2 [Implementing JWT security in Angular](#) ([Book Store v1.0](#))
 - 4.3 Implementing security in storage systems
 - 4.4 SSL
- 5 Middleware Message-Oriented MOM
 - 5.1 [Introduction](#) ([OPC UA Hangfire Server v1.x](#))
 - 5.2 Message-Brokers ([Web-API-Boilerplate-.NET-8](#))
 - 5.3 Open-Connectivity Unified Architecture OPC UA ([OPC UA Hangfire Server v1.x](#))
 - 5.4 SignalR ([Blazor-SignalR-Chat-.NET-8](#))
- 6 Storage
 - 6.1 In-memory Data Structure Storage
 - 6.1.1 Redis ([Book Store v2.x](#))
 - 6.2 NoSQL
 - 6.2.1 [MongoDB](#) ([Book Store v1.x](#))
 - 6.3 Relational
 - 6.3.1 M. SQL Server Full Text – Index ([Libreria-Search-v1.x](#))
 - 6.3.2 T-SQL ([OPC UA Hangfire Server v1.x](#))

- 6.3.3 PostgreSQL ([Books-Web-API-Docker-Postgresql-Redis-.NET-10](#))
- 6.4 Object Relational Mapping (ORM)
 - 6.4.1 Entity Framework ([OPC UA Hangfire Server v1.x](#))
 - 6.4.2 ADO.NET
- 7 Virtualisation technologies
 - 7.1 [VMware Workstation 25H2](#)
 - 7.2 Docker ([Book Store v1.x](#))
 - 7.2.1 Docker Compose ([Books-Web-API-Docker-Postgresql-Redis-.NET-10](#))
 - 7.3 Hyper-V
- 8 Automated Processes and Workflows
 - 8.1 Background Processing
 - 8.1.1 Hangfire Server ([OPC UA Hangfire Server v1.x](#))
- 9 Architectures, Designs and Patterns
 - 9.1 Clean Architecture
 - 9.1.1 Clean N-Tier Architecture ([Web-API-Boilerplate-.NET-8](#))
 - 9.2 Domain-Driven Architecture (DDA)
 - 9.3 [Hexagonal Architecture](#) ([Hexagonal-Arch-Web-API-.NET-10](#))
 - 9.4 Event-Driven Architecture EDA
- 10 ALC Methodologies and Procedures
 - 10.1 Waterfall ([OPC UA Hangfire Server v1.x](#)) *Technical document model*
 - 10.2 CI / CD
 - 10.2.1 [GitHub Actions for CI](#) ([Github-Actions-CI-CD-Example-WebAPI-.NET-8](#))
 - 10.2.2 [Watchtower for CD](#) ([Github-Actions-CI-CD-Example-WebAPI-.NET-8](#))
- 11 Unit Tests
 - 11.1 Mocks
 - 11.2 xUnit
- 12 Observability
 - 12.1 Prometheus ([Items-Web-API-React-19-PostgreSQL-.NET-8](#))
 - 12.2 Grafana ([Items-Web-API-React-19-PostgreSQL-.NET-8](#))
 - 12.3 cAdvisor ([Items-Web-API-React-19-PostgreSQL-.NET-8](#))
 - 12.4 Node Exporter ([Items-Web-API-React-19-PostgreSQL-.NET-8](#))
 - 12.5 Watchtower ([Github-Actions-CI-CD-Example-WebAPI-.NET-8](#))

13 Traditional .NET Applications

13.1 MVC

- 13.1.1 Razor ([*OPC UA Hangfire Server v1.x*](#)) Outdated, should be replaced with *Blazor*.

13.2 Web Forms

13.3 Windows Services

- 13.3.1 .NET Framework ([*Windows-Service-.Net-7*](#))
- 13.3.2 .NET ([*.NetFramework-Windows-Service-.NET-4.7*](#))

14 C# Corner

14.1 Fundamentals

14.2 .NET Task and Threads

- 14.2.1 Fundamentals ([*.NetFramework-Windows-Service-.NET-4.7*](#))
- 14.2.2 Types